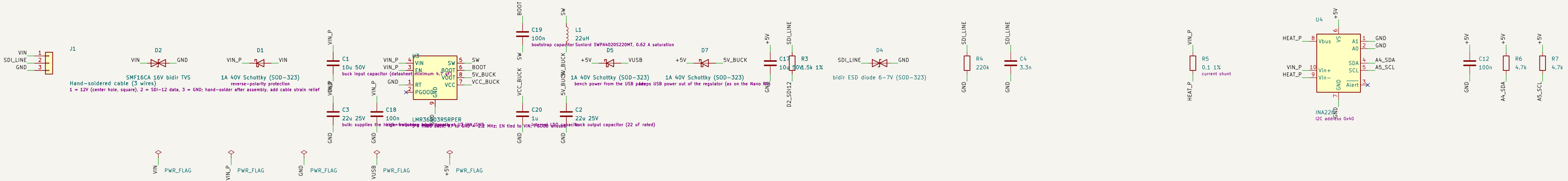


HP-SDI12-NANO / COMPACT FLAT – 11.5–14.4 V battery supply over the 3-wire SDI-12 cable. J1 has three bare cable solder holes (VIN, data, ground); no connector is fitted. 5 V comes from a buck converter (TI LMR36503R5, 5 V fixed, ~4 uA no-load, 2.2 MHz with RT tied to GND), which wastes several times less heat than a linear regulator.

SDI-12 DATA – meets the SDI-12 electrical numbers: R3 sets the 1–2 k transmit resistance (and limits fault current into pin D2), R4 the 160–360 k idle resistance, C4 holds the edge rate under 1.5 V/us. D4 clamps cable transients.

HEATER DRIVE + POWER MONITOR – PWM-capable D9/P303 drives Q1. Required firmware: 100 Hz, nominal 2 W average. 8–15 s; integrate stable ON power times actual ON time. Force D9 LOW at shutdown; stopping a timer is insufficient. R10 pulls an undriven gate LOW. Keep all three INA226 inputs on their dedicated Kelvin connections. Firmware is not included.



MICROCONTROLLER – Renesas RA4M1 (R7FA4M1AB3CFM), the Arduino UNO R4 / Nano R4 chip, wired exactly like the UNO R4 Minima so the stock board package and Arduino pin names apply (nets D2, D4, D9, A0–A5). Runs at 5 V. Support circuit (NMI, USB sense, VCC_USB, VCL) follows Arduino’s Nano R4 schematic. No 32.768 kHz crystal: the stock Arduino core clocks the CPU and USB from the internal HOCO and the RTC from LOCO.

